

CPP NOTES – DAY 32

Multiple Inheritance

Where a class can inherit from more than one classes. The constructors of inherited classes are called in the same order in which they are inherited. For example, in the following program, B's constructor is called before A's constructor. A class can be derived from more than one base class.

Multilevel Inheritance

A type of inheritance in C++ where one class inherits another class, which in turn is derived from another class. It is known as multi-level inheritance as there are more than one level of inheritance.

For example, if we take Grandfather as a base class then Father is the derived class that has features of Grandfather and then Child is the also derived class that is derived from the sub-class Father which inherits all the features of Father.

Eg: class A -> base

Class B: public A{}

Class C: public B{}

FYI

- The **order of inheritance** is importance(class C : public A,public B), here A 's constructor will be called first and then B.
- Person in one class and Address is another class, is it a good approach?
- Only public and protected can be inherited,no private.
- Why child class die first and parent later on?

Because the child depends on the parent. The child class might be using resources or methods from the parent. So, when it's time to clean things up:

- *You must release child-specific resources first (like files, memory, etc.)*
- *Only then can you safely tear down the parent part of the object.*

Destroying the parent first would leave the child operating on invalid or already-deleted parts of itself, which can lead to undefined behavior or errors.

- *When a base class is inherited with private access specifier, it will be kept private inside the derived class where base class cannot be accessed from another inherited class of derived class*
- *When base class is inherited with protected, it will be kept protected in derived class.*
- *Difference between single inheritance and hierarchical inheritance. Hierarchical is also a type of single inheritance but without having connection*